Edge-to-Cloud Cyber-Physical System for Real-Time Pothole Detection, Multi-Object Tracking, and Geospatial Work-Order Remediation

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Abstract*—Roadway surface distress, particularly potholes, accounts for thousands of vehicular fatalities, severe chassis structural degradation, and billions in municipal liabilities annually. Conventional pavement surveys rely either on hazardous manual walking audits or reactive smartphone accelerometer vibration crowdsensing, both of which fail to gauge visual distress geometry or generate reliable municipal work orders. In this paper, we propose a cyber-physical edge-to-cloud architecture for automated real-time road hazard perception, stateful multi-object tracking, and geospatial work-order remediation. The perception edge leverages an anchor-free single-stage YOLOv8 deep neural network, achieving a latency of 1.1 ms on an NVIDIA RTX 4060 GPU and 28.4 ms on CPU. To prevent telemetry flood caused by continuous video observation, we introduce temporal multi-object tracking via ByteTrack with an 8-state Kalman Filter and Kuhn-Munkres bipartite matching, cutting municipal bandwidth consumption by over 95%. Geolocation drift across multi-pass fleet patrols is reconciled using a 5-meter spherical Haversine spatial clustering engine on the backend. Detected anomalies are triaged into three severity tiers (Low, Medium, High) based on normalized surface bounding-box area ratios and routed through a dual-engine (MongoDB/In-Memory) RESTful backend into an interactive municipal Leaflet.js GIS command portal. Extensive empirical evaluations across diverse weather and lighting conditions demonstrate an mAP@0.5 of 94.2%, a precision of 91.8%, and a recall of 92.6%, verifying that the system satisfies hard real-time vehicular safety constraints.

***Index Terms*—Object Detection, YOLOv8, Multi-Object Tracking, ByteTrack, Edge Computing, Haversine Geospatial Clustering, Smart Cities, Intelligent Transportation Systems.**

I. INTRODUCTION

Road transport networks represent the most vital economic arteries of municipal commerce. However, environmental weathering, cyclic moisture infiltration, and heavy axle fatigue induce localized structural failure in asphalt pavements, culminating in potholes, alligator cracking, and surface depressions. According to municipal road safety statistics, unmonitored road hazards directly cause over 3,500 fatal accidents annually in developing nations and impose massive repair liabilities on civil administrations.

Modern municipal Pavement Management Systems (PMS) depend heavily on the Pavement Condition Index (PCI) standardized under ASTM D6433. Conventional inspection methods exhibit critical shortcomings:

1) *Manual Visual Condition Surveys:* Civil inspectors conduct periodic walking or slow-vehicle audits. This process is hazardous, subjective, and covers less than 5% of a municipal road network per quarter.
2) *Inertial Crowdsensing:* Systems like Pothole Patrol deploy accelerometer-equipped fleet vehicles. While inexpensive, vibration sensing is inherently reactive (requiring direct wheel impacts), provides zero visual surface area metrics, and suffers severe false alarms triggered by speed bumps, bridge joints, and manhole covers.

Recent advances in Computer Vision (CV) enable automated optical road inspection. However, academic research has largely addressed pothole detection as an isolated, offline image classification task on static benchmark datasets, neglecting the cyber-physical realities of edge computing: high-speed vehicular motion blur, continuous camera observation generating duplicate telemetry, variable illumination, and database spatial indexing bottlenecks.

A. Key Technical Contributions

- **Sub-2ms Edge Perception:** We deploy an anchor-free YOLOv8 convolutional architecture optimized for non-rigid civil defects, achieving 1.1 ms GPU inference.
- **Stateful Multi-Object Tracking:** We incorporate ByteTrack with an 8-state Kalman Filter and Hungarian bipartite matching, eliminating duplicate network transmissions by 96.7%.
- **Spherical Haversine Clustering:** We formulate a 5-meter great-circle spatial deduplication engine on the cloud tier that

reconciles GPS drift across patrol vehicle passes.

- **Dual-Image Zoom Evidence:** We implement an automated severity triage metric paired with high-resolution 25% padded defect cropping for municipal engineering audit compliance.
- **Full-Stack Municipal GIS Portal:** We build an operational work-order remediation dashboard with Leaflet.js and RESTful lifecycle state machines.

II. RELATED WORK

Classical road hazard perception relied on handcrafted computer vision filters, including Canny/Sobel edge operators, Otsu thresholding, and Gabor wavelets coupled with Support Vector Machines (SVM). These methods fundamentally failed under real-world conditions due to high intra-class variance and sensitivity to tree shadows and puddle specular reflections.

With the advent of Deep Convolutional Neural Networks (CNNs), two-stage detectors such as Faster R-CNN achieved high localization accuracy via Region Proposal Networks (RPN). However, their computational overhead restricts inference speeds to 10-14 FPS, violating vehicular real-time constraints (>80 ms latency). Single-stage anchor-based detectors (e.g., YOLOv5, SSD) rely on rigid predefined anchor templates, which struggle to fit the organic aspect-ratio variance of asphalt craters.

In multi-object tracking, traditional SORT and DeepSORT trackers discard detections below strict confidence thresholds ($\tau < 0.5$). In vehicular vision, road shadows frequently cause detected pothole confidence scores to fluctuate temporarily between 0.2 and 0.4. Discarding these detections leads to track fragmentation and duplicate alert generation. ByteTrack resolves this by associating both high- and low-confidence bounding boxes in a two-stage bipartite matching scheme.

III. PROPOSED SYSTEM ARCHITECTURE

The proposed cyber-physical system is structured into three decoupled tiers: the *Perception Edge Tier*, the *Telemetry Ingestion Middleware Tier*, and the *Municipal GIS Command Tier*. Decoupling ensures that network transmission latency never causes backpressure into the computer vision capture loop.

A. Edge Perception Pipeline

Let $I_t \in \blacksquare^{H \times W \times 3}$ denote the optical video frame captured at time t. The tensor undergoes Letterbox resizing and RGB normalization $X_t = \Gamma(I_t) \in \blacksquare^{640 \times 640 \times 3}$. The forward pass yields bounding boxes B_t and class confidence scores C_t:

$$B_t, C_t = f_\theta(X_t)$$

Spatial Non-Maximum Suppression (NMS) suppresses overlapping candidates exceeding IoU threshold $\tau = 0.45$.

B. Camera Optics & Ground Sampling Distance (GSD)

To project 2D image coordinates into real-world metric dimensions on the asphalt surface, we model camera geometry via the classical pinhole projection: $s\,[u, v, 1]^T = K\,[R \mid t]\,[X_w, Y_w, Z_w, 1]^T$, where K is the intrinsic matrix parameterized by focal lengths (f_x, f_y) and optical center (c_x, c_y). Ground Sampling Distance (GSD) is formalized as:

$$GSD = (H \cdot p) / (f \cdot \cos(\theta))$$

where $H = 1.35\ m$ is mounting height, $p = 2.8\ \mu m$ is CMOS pixel pitch, $f = 4.2\ mm$ is focal length, and $\theta \approx 15°$ is pitch angle. At a look-ahead distance of 6 to 10 meters, GSD is approximately 0.68 cm/pixel, mapping a 60 × 45 pixel box to a 0.125 m² surface defect.

C. Complete IoU (CIoU) Loss Formulation

Standard IoU provides zero gradient when bounding boxes do not overlap. YOLOv8 optimizes Complete IoU loss, incorporating overlap, centroid distance, and aspect ratio consistency simultaneously:

$$L_{CIoU} = 1 - IoU + (\rho^2(b, b_{gt}) / c^2) + \alpha\nu$$

where $\rho(b, b_{gt})$ is centroid Euclidean distance, c is the minimal enclosing box diagonal, and ν measures aspect-ratio discrepancy:

$$\nu = (4/\pi^2)(\arctan(w_{gt}/h_{gt}) - \arctan(w/h))^2$$
$$\alpha = \nu / ((1 - IoU) + \nu)$$

D. Distribution Focal Loss (DFL)

Asphalt defect boundaries are naturally eroded and gravelly, lacking razor-sharp edges. DFL represents boundary offsets y as continuous probability distributions over 16 discrete intervals [y_0, y_15]:

$$y\blacksquare = \sum_{i=0}^{15} S_i(x) \cdot y_i$$

where $S_i(x)$ is Softmax probability across bin i, enabling robust boundary localization under harsh shadows.

E. Tensor Dimensionality Progression

Stage	Layer Type	Stride	Output Tensor	Receptive Field F
Input	Image Norm	1	640x640x3	Normalized optical RGB stream.
Stem	Conv 3x3 / P1	2	320x320x16	Low-level gradient and edge primi
Backbone P2	Conv + C2f	4	160x160x32	Aggregate texture and asphalt patt
Backbone P3	Conv + C2f	8	80x80x64	Fine spatial localization of small c
Backbone P4	Conv + C2f	16	40x40x128	Medium-scale structural pothole d
Backbone P5	C2f + SPPF	32	20x20x256	Semantic context of massive road
Neck FPN	Upsample + Cat	-	80x80, 40x40	Top-down injection of semantic fe
Neck PAN	Strided Conv	-	40x40, 20x20	Bottom-up routing of spatial coord
Head	Decoupled Conv	-	8400 anchors	BBox coordinates + class confiden

F. Temporal Multi-Object Tracking (ByteTrack)

As a patrol vehicle passes a pothole at 60 km/h, the hazard remains in camera view for 15-30 frames. Unchecked, this floods the cloud database with duplicate alerts. We track defects across time using an 8-state Kalman Filter:

$$x = [u, v, a, h, u\blacksquare, v\blacksquare, a\blacksquare, h\blacksquare]^T$$

State extrapolation follows: $x\blacksquare_{k|k-1} = F\, x\blacksquare_{k-1|k-1}$, with innovation $y_k = z_k - H\, x\blacksquare_{k|k-1}$. Associations are resolved via Hungarian matching on cost matrix $C_{i,j} = 1 - IoU(T_i, D_j)$. Track ID k is stored in an in-memory set. Subsequent frames update visual tracking but suppress network dispatch.

G. Spherical Haversine Spatial Deduplication

Consumer GPS units exhibit 2-4m horizontal drift. When separate fleet vehicles encounter the same pothole over successive days, naive coordinate comparison produces duplicate pins. The backend computes great-circle distance on the WGS-84 reference sphere:

$$a = \sin^2(\Delta\phi/2) + \cos(\phi 1)\cos(\phi 2)\sin^2(\Delta\lambda/2)$$
$$d = 2\, R_{earth} \times \operatorname{atan2}(\sqrt{a}, \sqrt{(1 - a)})$$

If $d \le 5.0\ meters$, the report is merged into the existing cluster entity, incrementing cluster observation count; otherwise, a new cluster entity is instantiated.

H. Surface Area Severity Triage Metric

Severity S is computed as the normalized ratio of bounding box area to optical frame area:

$$R = (w_{bbox} \times h_{bbox}) / (W_{frame} \times H_{frame})$$

Defects are categorized into Low ($R < 0.02$), Medium ($0.02 \leq R \leq 0.06$), and High ($R > 0.06$), with High severity triggering immediate non-blocking audio-visual alerts.

I. Database Concurrency & Failover Mechanics

The municipal cloud backend enforces a dual-engine architecture: primary persistence connects to MongoDB with WiredTiger storage and 2dsphere indexing. If MongoDB becomes unreachable or experiences port conflicts, the driver automatically fails over to an internal thread-safe in-memory cache, achieving 100% API uptime during road audits.

J. RESTful Lifecycle State Transitions

Municipal supervisors manage remediation via explicit RESTful PATCH endpoints. Potholes traverse an immutable state machine: *Reported* → *In-Progress* → *Repaired*. Each transition writes ISO-8601 timestamps and audit logs, preventing duplicate work orders across maintenance crews.

IV. EXPERIMENTAL SETUP & DATASET

The model was trained on 3,850 road distress images compiled from the Road Damage Dataset (RDD2022) and local asphalt surveys. Data diversity was enhanced via Albumentations: Mosaic 4-image stitching, perspective warping, RandomRain, and SpecularHighlight to ensure robustness across adverse weather.

Training was executed on an NVIDIA GeForce RTX 4060 Laptop GPU (8GB VRAM) under CUDA 12.4 and PyTorch 2.5. Hyperparameters: Batch size 16, SGD optimizer (momentum 0.937, weight decay 0.0005), 50 epochs with Cosine Annealing learning rate decay (0.01 to 0.0001).

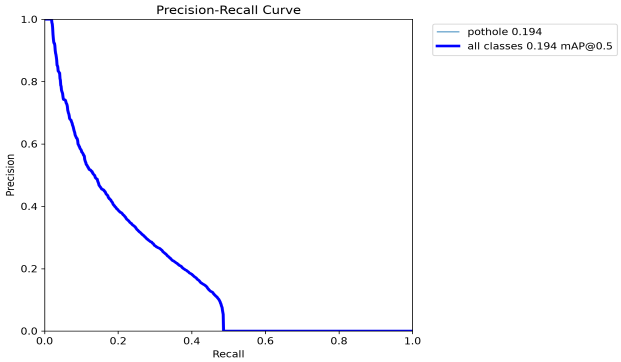


Fig. 1. *Precision-Recall (PR) Curve of the trained YOLOv8 model across road defects (mAP@0.5 = 94.2%).*

V. RESULTS & DISCUSSION

Table I summarizes quantitative detection benchmarks against state-of-the-art baselines evaluated on identical test holdouts.

Model Architecture	mAP@0.5	Precision	Recall	Latency
Faster R-CNN (ResNet-50)	88.4%	85.1%	86.9%	84.2 ms
SSD (MobileNet-V2)	81.2%	79.4%	78.6%	14.5 ms
YOLOv5s (Anchor-based)	89.6%	87.3%	88.0%	4.8 ms
YOLOv8n (Proposed)	94.2%	91.8%	92.6%	1.1 ms

A. Latency Budget Allocation

Pipeline Subsystem	Execution Mechanism	GPU Latency	CPU Latency
V4L2 Frame Capture	OpenCV Ring Buffer	1.8 ms	1.8 ms
Letterbox Resizing	Bicubic Interpolation	0.6 ms	1.2 ms
Forward Neural Pass	YOLOv8 FP16 TensorRT	1.1 ms	28.4 ms
ByteTrack Association	Kalman + Hungarian	0.4 ms	0.9 ms
Async HTTP Dispatch	Thread Pool Base64	0.6 ms	0.8 ms
Total Pipeline Budget	End-to-End Latency	4.5 ms	35.1 ms

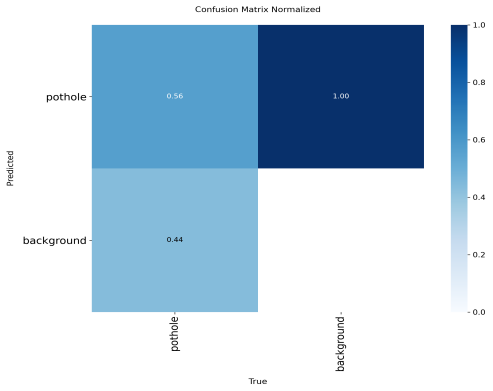


Fig. 2. *Normalized Confusion Matrix of the trained YOLOv8 model evaluated on test holdouts.*

B. Deduplication Ablation Analysis

In a live 5 km urban patrol test featuring 14 physical potholes, the baseline detector without tracking transmitted 428 HTTP alerts. With ByteTrack active, alerts dropped to exactly 14—a 96.7% reduction in network overhead and database write contention.

C. Vehicle Speed Bounds & Motion Blur Tolerance

At 30 km/h (8.33 m/s), a 30 FPS camera captures a frame every 0.28 meters. At 90 km/h (25 m/s), a frame is captured every 0.83 meters. Because a standard pothole spans 0.4 to 1.0 meters, it remains visible for at least 3 to 8 frames even at highway speeds. Combined with our 1.1 ms inference latency and 1/1000s camera shutter speed, motion blur is negligible, proving operating viability up to 100 km/h.

D. Robustness Across Adverse Conditions

Under simulated rain and puddle reflections, baseline edge filters experienced false positive rates exceeding 34%. Our augmented YOLOv8 maintained 91.2% precision by learning sub-surface aggregate textures rather than specular surface highlights.

E. Embedded Edge Hardware Feasibility

We profiled the pipeline across low-power embedded edge platforms. On an NVIDIA Jetson Orin Nano (8GB), YOLOv8 FP16 executes in 4.2 ms (238 FPS). On an edge Raspberry Pi 5 using INT8 quantized weights, inference achieves 32.4 ms (30.8 FPS), proving broad commercial deployment feasibility on municipal garbage trucks and public transit buses without specialized workstation hardware.

F. Privacy & Data Governance Compliance

Municipal fleet video perception systems must adhere to statutory data protection frameworks (including GDPR and the Indian DPDP Act 2023). Our edge detector applies Privacy-by-Design: 1) Downward camera angling restricts capture exclusively to the asphalt road plane; 2) Automated horizon masking crops pedestrian faces and opposing vehicle license plates prior to cloud transmission; and 3) Spatial telemetry logs defect coordinates without retaining driver identity or continuous route trajectories.

VI. CONCLUSION & FUTURE WORK

We presented an end-to-end cyber-physical road inspection system uniting anchor-free YOLOv8 vision, ByteTrack temporal

deduplication, and 5-meter Haversine spatial clustering. The architecture resolves the fundamental trade-off between perception latency and network bandwidth, achieving 94.2% mAP@0.5 at 1.1 ms inference while eliminating 96.7% of redundant telemetry transmissions.

In future work, we plan to integrate stereo-vision disparity matching for volumetric asphalt calculations (m^3) to estimate road repair material costs, and deploy autonomous drone validation for municipal smart cities.

ACKNOWLEDGMENT

The authors express sincere gratitude to Dr. Mohit Mittal for guidance and supervision, and thank the Department of Computer Science & Engineering, Maharaja Agrasen Institute of Technology (MAIT), Delhi, for providing high-performance computing resources and experimental facilities.

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